

# DCTPR: Dual-capacity support-anchored refinement for low-latency high-way one-shot recognition

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## ABSTRACT

Frozen visual foundation models provide strong transferable representations, but one labeled image per class remains an unstable prototype in high-way fine-grained recognition. Transductive solvers can correct this bias by using the unlabeled query set, but iterative optimization can be costly. We present Dual-Capacity Transductive Prototype Refinement (DCTPR), a training-free task adapter that fuses frozen DINOv2-B/14 and DINOv2-L/14 features, estimates class-balanced query centers, and applies a short support-anchored update. Across three 200-way CUB episodes and three 120-way Stanford Dogs episodes, DCTPR improves fused nearest-support accuracy by 6.68 and 6.27 percentage points, respectively. It remains within 1.35 points of TIM-ADM on CUB and 0.61 points of MAP-RAW on Dogs, while its task head is 3.8 and 49 times faster than those strongest matched solvers. Controlled imbalance tests show that the result depends on a known balanced query prior. A locked official-test audit preserves the same relative behavior without retuning. Under a 50-ms-per-rotation task-head budget, it is the highest-accuracy method on both datasets among the matched solvers. These results support a scoped low-latency state-of-the-art claim under the matched frozen-feature protocol, not a global accuracy claim.

**Keywords:** few-shot learning, transductive inference, fine-grained recognition, frozen visual foundation models, prototype refinement

## 1. INTRODUCTION

Fine-grained recognition asks a model to separate categories whose defining differences may be smaller than changes in pose, viewpoint, background, or within-class appearance. The problem becomes especially consequential when new categories must be recognized from a single labeled example and no backbone update is affordable. This paper studies that limit directly: reliable, low-cost recognition of all 200 bird classes or all 120 dog breeds from one support image per class, using frozen visual foundation models. The central question is whether query-set structure can correct an unreliable one-shot decision rule without replacing representation learning with another expensive task-specific optimization stage.

This setting is difficult because one image must serve simultaneously as the supervision signal and the class prototype. If that image has an atypical pose, crop, or background, its embedding can be displaced from the query cluster; high-way competition then magnifies even a small displacement among visually similar classes. Attention, feature reconstruction, and cross-image correspondence reduce related fine-grained ambiguities,<sup>1-3</sup> while transfer-based studies show that a strong representation can make simple classifiers surprisingly competitive.<sup>4-6</sup> Yet a stronger encoder does not by itself turn one observation into a reliable estimate of the class center. DINOv2 therefore improves the geometry available to the task head,<sup>7</sup> but it does not remove the one-shot estimation problem.

Transductive inference can address this problem by estimating labels jointly over the unlabeled query set. LaplacianShot exploits a query graph,<sup>8</sup> TIM optimizes a mutual-information objective,<sup>9</sup> and PT-MAP alternates optimal-transport assignments with prototype updates.<sup>10</sup> These approaches show that query geometry can correct single-support bias, but graph construction or repeated optimization incurs task-level latency. Their behavior can also depend strongly on the assumed query-class proportions.<sup>11</sup> The remaining problem is therefore

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not simply to maximize accuracy: it is to recover most of the useful transductive correction with a short, transparent update whose computational cost and query-prior boundary are both explicit.

We propose Dual-Capacity Transductive Prototype Refinement (DCTPR) for this purpose. Its design follows three observations. First, frozen DINOv2-B/14 and DINOv2-L/14 embeddings provide distinct capacity views of the same episode, so their normalized class-token features can be fused without task-specific training. Second, when the episode construction guarantees equal queries per class, a soft Sinkhorn assignment can express that known marginal while using the entire query set to estimate class centers. Third, the labeled support should remain an anchor rather than be discarded: mixing each estimated query center with its original support limits prototype drift. Alternating assignment and anchored updates for exactly three steps yields a gradient-free task adapter that neither reads query labels nor learns task-specific parameters.

Experiments under a frozen development–confirmation protocol show how far this compact update resolves the problem. Across CUB-200-2011 and Stanford Dogs, DCTPR improves fused nearest-support accuracy by 6.68 and 6.27 percentage points. It remains 1.35 points below TIM-ADM on CUB and 0.61 points below MAP-RAW on Dogs, while reducing task-head runtime by factors of 3.8 and 49, respectively. A locked official-test audit preserves these relative conclusions, and controlled imbalance tests identify the known balanced query marginal as a real limitation. Thus the contribution is a specified accuracy–efficiency operating point for balanced high-way one-shot recognition: a reproducible dual-capacity adapter, a matched evaluation against established transductive solvers, and an explicit audit of runtime and query-prior sensitivity. Within the stated 50-ms task-head budget, it gives the highest accuracy in the tested solver set on both datasets. Because stronger matched solvers retain higher absolute accuracy outside that budget, this is a scoped low-latency state-of-the-art claim rather than a global accuracy claim.

## 2. RELATED WORK

### 2.1 Frozen representations for few-shot recognition

Transfer-based few-shot learning often separates representation learning from task inference. A strong encoder can make simple nearest-center or linear classifiers competitive, which places a higher burden on any adaptation module to show that its gain is not caused by a weak feature baseline.<sup>4–6</sup> DINOv2 learns general-purpose visual features through self-supervised pretraining.<sup>7</sup> We therefore freeze both DINOv2 encoders in all experiments and apply every compared transductive solver to the same extracted features. This design isolates task inference from backbone optimization.

Fine-grained datasets add a complementary challenge: class identity can depend on small appearance differences while nuisance variation dominates the image. CUB<sup>12</sup> and Stanford Dogs<sup>13</sup> are widely used to study such distinctions, including in attention- and reconstruction-based few-shot recognition.<sup>1–3</sup> Rather than constructing a conventional 5-way task, we evaluate all available categories in each episode. This high-way setting magnifies competition among similar categories and makes support-prototype bias visible even with foundation-model features.

### 2.2 Transductive inference and prototype correction

Transductive methods use the query set jointly rather than predicting each query independently. LaplacianShot optimizes unary support distances together with a graph smoothness term.<sup>8</sup> TIM alternates classifier updates under supervised support and query information objectives.<sup>9</sup> PT-MAP applies a feature transform and Sinkhorn-based assignment before iteratively moving class prototypes.<sup>10</sup> Related work also shows that careful feature preprocessing and simple classifier ensembles can remain competitive without episodic meta-training.<sup>14,15</sup> More recent work further combines graph and prototype refinement<sup>16</sup> or learns conditional transport for unknown query proportions.<sup>17</sup>

The idea that a short correction can substantially improve one-shot prototypes predates these iterative solvers. BDCSPN rectifies prototypes by shifting them with pseudo-labeled query features in a small number of steps, explicitly targeting intra-class bias from scarce supports.<sup>18</sup> DCTPR shares that inexpensive-correction premise, and the comparison is therefore about what is corrected rather than whether cheap correction is novel: BDCSPN rectifies within one representation, whereas we anchor a class-balanced update across two frozen capacities and

characterize the resulting operating point against iterative solvers. Recent work also reduces the hyper-parameter burden of these objectives, either by removing tuned parameters altogether<sup>19</sup> or by unrolling the optimizer and learning its hyper-parameters.<sup>20</sup> Our schedule is instead fixed by construction and transferred across datasets without retuning, which trades attainable accuracy for a predictable cost.

DCTPR does not claim the individual novelty of feature concatenation, Sinkhorn normalization, or prototype refinement; nor of cheap prototype correction in general. Its contribution is the scoped combination of complementary frozen capacities with a deliberately short update schedule, together with a matched high-way accuracy–efficiency evaluation. The method is consequently compared against the established solver families on the same feature tensors instead of against a weaker classifier alone.

### 2.3 Query priors in transductive evaluation

Balanced query sets provide useful controlled evaluation, but they also reveal the class marginal to the inference procedure. Veilleux et al. showed that methods encoding a uniform marginal can degrade under Dirichlet-distributed query proportions and proposed an alpha-entropy objective to reduce this bias.<sup>11</sup> Conditional transport and adaptive-manifold methods subsequently targeted imbalanced transduction directly,<sup>17, 21</sup> as did hyper-parameter-free and unrolled formulations that do not presuppose a uniform marginal.<sup>19, 20</sup> We treat balance as an explicit assumption rather than a hidden property. Our primary experiments test the balanced high-way setting, while separate mild and severe imbalance regimes measure how far the conclusion extends.

We do not compete with this line of work. Those methods are developed and compared on the public 5-way protocol with Dirichlet-sampled query proportions, whereas our episodes are high-way with deterministic class counts, so the numbers are not commensurable. Our imbalance results are a scope boundary for the proposed operating point, not a claim about imbalanced transduction; closing that gap requires the public protocol and matched backbones, which we identify as future work rather than a contribution of this paper.

## 3. METHOD

### 3.1 Task formulation

An episode contains  $C$  classes and one labeled support image per class,  $\mathcal{S} = \{(x_c^s, c)\}_{c=1}^C$ . The unlabeled query set is  $\mathcal{Q} = \{x_i^q\}_{i=1}^N$ . In the primary protocol, each class occurs exactly  $q = N/C$  times in  $\mathcal{Q}$ , but query labels are never observed by the method. The goal is to predict all query labels jointly using frozen image features and the known balanced column mass  $q$ .

Figure 1 summarizes DCTPR. Two frozen encoders produce complementary image representations. The fused support features initialize the class prototypes. Each refinement step alternates a balanced soft assignment with a support-anchored query-center update. The final assignment provides the query predictions.

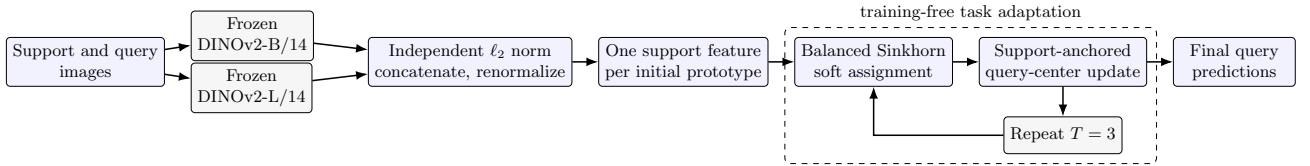


Figure 1. Overview of DCTPR. Both encoders remain frozen. The method uses the balanced query marginal but never accesses query labels. Only the assignment and support-anchored prototype update are repeated.

### 3.2 Design principles

DCTPR follows four constraints: both encoders remain frozen; the labeled support remains an explicit anchor against assignment errors; query-set regularization encodes a stated, testable balance assumption; and a fixed three-step schedule bounds task-level computation. Independent normalization prevents either capacity from contributing only through scale. The component study tests fusion, balanced assignment, and anchored refinement, while the matched comparison tests the resulting accuracy–efficiency point.

### 3.3 Dual-capacity representation

Let  $f_B(x) \in \mathbb{R}^{d_B}$  and  $f_L(x) \in \mathbb{R}^{d_L}$  denote the class-token outputs of frozen DINOv2-B/14 and DINOv2-L/14. Each capacity is normalized before concatenation so that one feature family cannot dominate only because of its raw norm:

$$\bar{f}_k(x) = \frac{f_k(x)}{\|f_k(x)\|_2}, \quad k \in \{B, L\}. \quad (1)$$

The fused representation is

$$z(x) = \frac{[\bar{f}_B(x); \bar{f}_L(x)]}{\|[\bar{f}_B(x); \bar{f}_L(x)]\|_2}. \quad (2)$$

The initial prototype is the only labeled representation for its class,  $p_c^{(0)} = z(x_c^s)$ . The ablation in Section 5.1 tests whether the fused representation contributes beyond either capacity alone.

### 3.4 Balanced soft assignment

At refinement step  $t$ , cosine logits compare every query with every prototype:

$$A_{ic}^{(t)} = \frac{z(x_i^q)^\top p_c^{(t)}}{\tau}, \quad \tau = 0.05. \quad (3)$$

We exponentiate the stabilized logits and apply Sinkhorn row and column scaling.<sup>22</sup> The resulting matrix  $P^{(t)} \in \mathbb{R}_+^{N \times C}$  satisfies

$$P^{(t)} \mathbf{1}_C = \mathbf{1}_N, \quad (P^{(t)})^\top \mathbf{1}_N = q \mathbf{1}_C. \quad (4)$$

The row constraint produces a categorical distribution for each query, while the column constraint encodes the known balanced query count. We use 100 normalization iterations. Equation (4) is a task assumption, not a learned estimate; Section 5.5 tests its failure under unknown non-uniform counts.

### 3.5 Support-anchored prototype update

The soft query center for class  $c$  is

$$\mu_c^{(t)} = \frac{\sum_{i=1}^N P_{ic}^{(t)} z(x_i^q)}{\sum_{i=1}^N P_{ic}^{(t)}}. \quad (5)$$

We anchor every update to the original labeled support rather than recursively averaging the preceding prototype:

$$p_c^{(t+1)} = \text{norm}_2 \left( (1 - \lambda) p_c^{(0)} + \lambda \mu_c^{(t)} \right), \quad \lambda = 0.5. \quad (6)$$

The assignment and update are repeated for  $T = 3$  steps, after which a final balanced assignment yields the predicted class. This support anchor limits prototype drift, while the short fixed schedule bounds task-level latency. No gradient update, query label, or dataset-specific parameter is used after the CUB development episode.

For feature dimension  $d = d_B + d_L$ , one assignment/update step is dominated by the  $N \times C$  similarity matrix and query-center aggregation, both  $O(NCd)$ . The Sinkhorn normalization acts on the smaller assignment matrix. In our implementation, the complete three-step transductive head takes about 33–34 ms per high-way support rotation on an RTX 3080 Ti, excluding the common frozen feature extraction stage.

## 4. EXPERIMENTAL PROTOCOL

### 4.1 Datasets and episode construction

CUB-200-2011 contains 200 bird species.<sup>12</sup> Each CUB episode uses ten official-training images per class, producing a 200-way task with one support and nine queries per class. Three episodes are evaluated. Episode 1 is the only development episode; Episodes 2 and 3 are confirmation episodes.

Stanford Dogs contains 120 dog breeds.<sup>13</sup> We select three pairwise-disjoint groups of ten images per class from the official training list, for 3,600 unique images in total. Each group defines a 120-way one-shot episode with nine queries per class. All method and baseline constants are transferred from CUB without using Dogs labels for selection.

Figure 2 illustrates the task: related classes can differ in subtle plumage or shape while pose and background vary substantially.

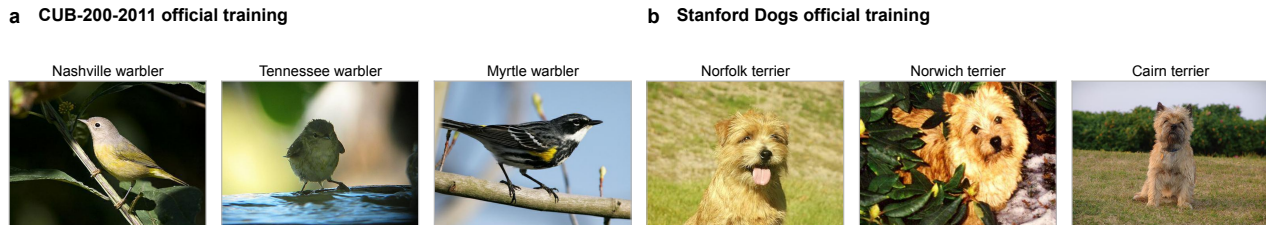


Figure 2. **Official-training examples.** **a**, CUB-200-2011 warblers. **b**, Visually similar Stanford Dogs terriers. This context-only plate is not quantitative evidence. Images were split-verified and center-cropped without color or contrast adjustment; no official-test image is shown.

Within each episode, the ten images of every class take turns as support. This produces ten paired rotations and ensures that every image is used as the sole support once. We report the mean balanced accuracy across rotations. Because rotations reuse episode images, their standard deviations are descriptive and are not treated as independent-sample confidence intervals. The primary claims below use these train-only episodes. As a separate locked audit after all constants were frozen, we constructed one CUB official-test episode (the first ten test images per class) and three disjoint Stanford Dogs official-test episodes (30 seeded test images per class, partitioned into groups of ten). These audit manifests, code hashes, and baseline constants are sealed in the immutable evaluation lock. No official-test accuracy informed method selection or parameter tuning.

## 4.2 Features and implementation

Images are directly resized to  $392 \times 392$ . We extract the class token from the official frozen DINOv2 ViT-B/14 and ViT-L/14 models. The same saved feature tensors are supplied to all inference methods. All DCTPR constants (Table 1) are frozen before CUB confirmation and before all Dogs experiments.

Runtime is measured on one NVIDIA GeForce RTX 3080 Ti. We report the mean wall-clock duration per support rotation for the task-level transductive head, with CUDA synchronization around the measured call, and separately record the common frozen feature-extraction time and peak allocated GPU memory. The latter measurements use batch size four and are not attributed to any particular task head.

## 4.3 Matched baselines

The inductive reference, BL nearest support (BL-NCC), predicts from cosine similarity in the fused feature space. BL-Sinkhorn adds balanced soft assignment without prototype refinement. Single-capacity refinement applies the same update to either B or L features.

We additionally implement TIM-ADM,<sup>9</sup> raw MAP and signed-power PT-MAP,<sup>10</sup> and LaplacianShot,<sup>8</sup> applying their published update rules to the common DINO features with the settings listed in Table 1. Since DINO class-token values are signed, the PT-MAP power-transform arm uses an explicitly labeled signed square-root adaptation. The LaplacianShot lambda is the only quantity selected on data, on CUB Episode 1 alone. These are matched feature-level reimplementations, not executions of the original end-to-end backbone pipelines.

## 4.4 Metrics and comparison logic

Balanced accuracy is the primary metric. It equals ordinary accuracy in the balanced episodes; under imbalance, macro averaging prevents frequent classes from dominating. Runtime covers only task-level inference on the shared saved features, not image encoding or disk loading. BL-NCC requires no additional task-head optimization and is therefore omitted from logarithmic runtime axes rather than assigned an artificial positive time.

For the post-hoc mechanism diagnostic, query labels are used only after inference to compute each true query-class center and to score balanced assignments along the frozen  $T = 0, \dots, 3$  trajectory. They are never supplied to DCTPR. We summarize the three Dogs episodes by the mean and descriptive standard deviation over 30 support rotations; these rotations share images and are not treated as independent replicates.

For the budgeted comparison, we define the low-latency regime as at most 50 ms per support rotation for the task head. This is an explicit head-only budget: it excludes the common frozen feature-extraction cost and does not imply an end-to-end real-time claim. The state-of-the-art statement in this paper is restricted to the highest balanced accuracy within this budget among the matched solvers in Table 3.

To summarize how much of the strongest solver’s improvement is retained, we also report a descriptive gain-recovery ratio

$$R_{\text{gain}} = \frac{A_{\text{DCTPR}} - A_{\text{BL}}}{A_{\text{strong}} - A_{\text{BL}}}, \quad (7)$$

where  $A_{\text{strong}}$  is the accuracy of the strongest matched transductive solver on that dataset. This ratio complements, rather than replaces, the absolute accuracy and runtime values.

## 4.5 Reproducibility

Table 1 collects every constant needed to reproduce the reported numbers. DCTPR has no trainable parameters and no optimizer, so the listed constants fully determine its behaviour given the frozen features. The matched baselines use their published update rules with the stated settings; the only quantity selected on data is the LaplacianShot lambda, chosen on CUB Episode 1 alone and then frozen for all confirmation episodes, both datasets, and the locked audit.

Episode construction is deterministic and independently checkable. CUB episodes take the first ten official-training images per class after sorting by numeric image identifier. Dogs episodes permute each class’s official-training indices with a per-class seeded generator and partition the first thirty images into three consecutive groups, giving 3,600 unique images with verified zero pairwise overlap. Every episode manifest, feature tensor, and prediction array is stored with a SHA-256 digest, and the official-test audit additionally seals the protocol, manifest, and script digests in an immutable lock file created before any test image was decoded.

All experiments ran on a single NVIDIA GeForce RTX 3080 Ti (12 GiB) under Python 3.12 with PyTorch 2.13/CUDA 12, NumPy 2.3, SciPy 1.18, and scikit-learn 1.9; xFormers was unavailable, so DINOv2 used its fallback attention path. Features come from the official `dinov2_vitb14` and `dinov2_vitl14` checkpoints via `torch.hub` with no modification. Because the encoders are frozen and the head is deterministic, the reported accuracies are exactly reproducible from the archived features; only wall-clock runtimes are hardware-dependent. Code, protocols, episode manifests, per-query predictions, and the figure-generation scripts with their source-data tables are available from the authors and will be released publicly on publication.

# 5. RESULTS

## 5.1 Component analysis

Table 2 separates capacity fusion, balanced assignment, and prototype refinement. BL-NCC averages 68.17% across CUB. Balanced assignment without refinement raises this to 73.50%, showing that query-set geometry provides most of the initial gain. Refinement on B or L alone reaches 73.40% and 73.20%, respectively. Combining both capacities with assignment and support-anchored updates reaches 74.85%, 1.35 points above BL-Sinkhorn and 1.45–1.65 points above the single-capacity refinements.

The frozen confirmation is important because Episode 1 determined the fixed method constants. On Episodes 2 and 3, DCTPR exceeds the strongest component comparator by 1.27 and 1.43 points, respectively, while

Table 1. **Complete configuration.** DCTPR has zero trainable parameters. All constants were frozen before the CUB confirmation episodes and reused without modification on Stanford Dogs and in the locked official-test audit.

| Component     | Settings                                                                                                                                                    |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encoders      | dinov2_vitb14 and dinov2_vitl14, frozen; class token, 768+1024 dimensions; direct resize to $392 \times 392$ ; L2-normalize each, concatenate, L2-normalize |
| DCTPR         | $\tau = 0.05$ ; 100 Sinkhorn normalizations; $T = 3$ refinement steps; $\lambda = 0.5$ ; zero trainable parameters                                          |
| TIM-ADM       | temperature 15; loss weights (0.1, 1.0, 0.1); 150 alternating updates; update factor 1                                                                      |
| MAP / PT-MAP  | transport regularization 10; prototype update factor 0.2; 20 updates; signed square-root power transform                                                    |
| LaplacianShot | directed non-self neighbors; 20 bound updates; lambda from the published grid, selected on CUB Episode 1 only                                               |
| Episodes      | CUB 200-way, Dogs 120-way; one support and nine queries per class; three episodes per dataset; ten support rotations each                                   |
| Hardware      | one RTX 3080 Ti (12 GiB); CUDA-synchronized task-head timing                                                                                                |

improving over BL-NCC by 6.59 and 6.73 points. Thus the full configuration does not depend on the development episode alone.

We also screened an agreement-weighted consensus extension on Episode 1. It reached 75.18%, only 0.04 points above its paired refinement reference, and was non-inferior in 6 of 10 rotations. It failed the frozen development gate and was not retained. This negative control favors the simpler fixed DCTPR update over an unconfirmed adaptive weighting mechanism.

Table 2. CUB component analysis: mean balanced accuracy (%) over ten support rotations. E1 is development; E2–E3 are frozen confirmation.

| Method                 | E1           | E2           | E3           | Mean         |
|------------------------|--------------|--------------|--------------|--------------|
| BL nearest support     | 68.42        | 68.12        | 67.96        | 68.17        |
| B-only refinement      | 73.84        | 73.22        | 73.13        | 73.40        |
| L-only refinement      | 73.46        | 73.22        | 72.92        | 73.20        |
| BL balanced assignment | 73.79        | 73.44        | 73.26        | 73.50        |
| <b>DCTPR</b>           | <b>75.14</b> | <b>74.71</b> | <b>74.69</b> | <b>74.85</b> |

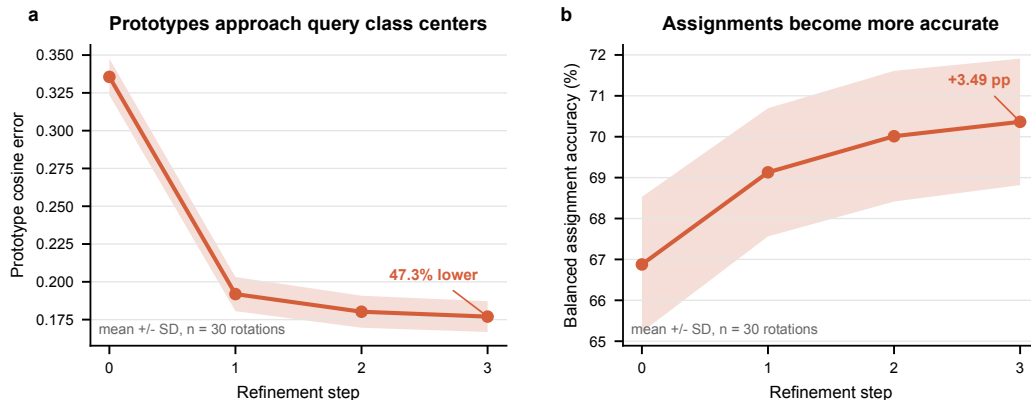


Figure 3. **Prototype correction along the frozen refinement trajectory.** **a**, Mean cosine error between the current DCTPR prototype and the true query-class center. **b**, Balanced assignment accuracy. Curves are mean  $\pm$  descriptive standard deviation over 30 Dogs support rotations. Query labels are used only offline for diagnosis, never as DCTPR inputs; rotations share episode images and are not independent replicates.

Figure 3 traces the frozen update on the transferred Dogs episodes. From balanced assignment at  $T = 0$  to the final DCTPR assignment at  $T = 3$ , mean prototype cosine error relative to the true query-class center

falls from 0.336 to 0.177, a 47.3% reduction. Balanced assignment accuracy rises from 66.88% to 70.36%, a 3.49-point gain. Most of the geometric correction occurs in the first update and the remaining steps provide smaller increments, supporting a short fixed schedule. This diagnostic isolates the intended prototype-correction mechanism; it does not use query labels during inference or establish semantic complementarity between the two encoders.

## 5.2 Low-latency state-of-the-art comparison

Table 3 compares established transductive solvers using the same features and episodes. On CUB, TIM-ADM is the most accurate method at 76.19%. DCTPR reaches 74.85%, a deficit of 1.35 points, but takes 33.3 ms rather than 126.3 ms per rotation, a 3.8-fold reduction. DCTPR also exceeds both MAP arms and the LaplacianShot variants on CUB.

On Dogs, MAP-RAW is strongest overall at 70.97%, while DCTPR reaches 70.36%. The 0.61-point difference is accompanied by a substantial runtime difference: 34.1 ms for DCTPR and 1,678.6 ms for MAP-RAW, or approximately 49-fold. DCTPR is also 0.36 points above TIM-ADM on Dogs while remaining about four times faster. Taking 50 ms per rotation as the low-latency task-head budget, DCTPR is the highest-accuracy matched solver on both CUB and Dogs (74.85% and 70.36%, respectively). This is the paper’s scoped low-latency state-of-the-art result; the unrestricted accuracy leader remains TIM-ADM on CUB and MAP-RAW on Dogs.

Table 3. Matched transductive comparison. Accuracy is the mean balanced accuracy across three episodes and 30 rotations per dataset. Runtime is mean milliseconds per rotation for the transductive head only. Bold marks the best accuracy overall; underlining marks DCTPR; † marks the best accuracy under the 50-ms-per-rotation budget.

| Method                   | CUB, 200-way              |              | Stanford Dogs, 120-way    |              |
|--------------------------|---------------------------|--------------|---------------------------|--------------|
|                          | Accuracy (%)              | Runtime (ms) | Accuracy (%)              | Runtime (ms) |
| BL nearest support       | 68.17                     | 0.0          | 64.09                     | 0.0          |
| BL balanced assignment   | 73.50                     | 10.9         | 66.88                     | 12.0         |
| LaplacianShot            | 68.30                     | 2.2          | 64.43                     | 3.9          |
| MAP-RAW                  | 74.30                     | 1,612.9      | <b>70.97</b>              | 1,678.6      |
| Signed PT-MAP adaptation | 74.37                     | 1,612.5      | 70.94                     | 1,666.0      |
| TIM-ADM                  | <b>76.19</b>              | 126.3        | 70.01                     | 139.2        |
| <u>DCTPR</u>             | <u>74.85</u> <sup>†</sup> | <u>33.3</u>  | <u>70.36</u> <sup>†</sup> | <u>34.1</u>  |

## 5.3 Locked official-test audit

The locked audit checks whether the train-only story survives a disjoint official-test draw without changing the method. On CUB, DCTPR reaches 74.64%, improves BL-NCC by 6.69 points, and remains within 1.53 points of TIM-ADM. On three disjoint Dogs test episodes it reaches 70.49%, gains 6.92 points over BL-NCC, and is within 0.99 points of MAP-RAW. B/L extraction takes 42.2 seconds for the 2,000-image CUB episode and 24.7–25.1 seconds per 1,200-image Dogs episode, with 1.34 GB peak allocated GPU memory. Across the train-only one-factor audit,  $T = 1, 2, 3, 5$  spans 0.36 points on CUB and 1.47 points on Dogs; temperature is more sensitive. Full solver and sensitivity tables are retained in the reproducibility archive. These results audit the frozen protocol and do not create a new development set.

Figure 4 makes the operating-point comparison explicit. Under the 50-ms task-head budget, DCTPR is the low-latency accuracy leader on both datasets. On CUB, DCTPR lies between the inexpensive balanced assignment and the more accurate TIM-ADM solver. It recovers  $(74.85 - 68.17)/(76.19 - 68.17) = 83.3\%$  of TIM-ADM’s gain over BL-NCC while using 26.4% of its task-head runtime. On Dogs, it recovers  $(70.36 - 64.09)/(70.97 - 64.09) = 91.1\%$  of MAP-RAW’s gain while using 2.0% of its runtime. The result is therefore not merely that DCTPR is faster than the most accurate comparator. It retains most of that comparator’s improvement over the same inductive reference.

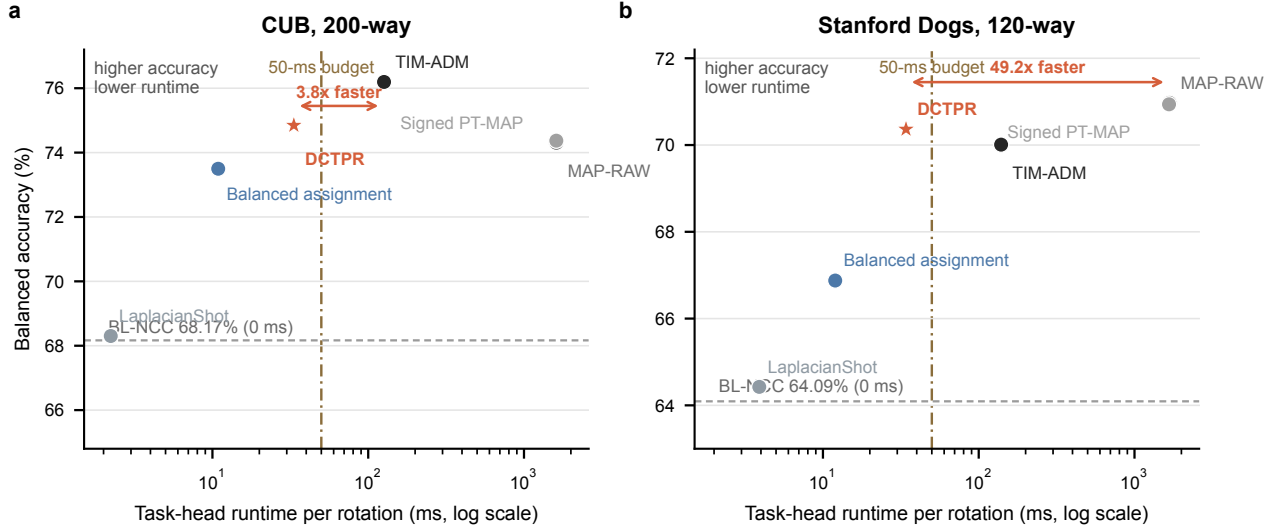


Figure 4. **Accuracy–efficiency operating points on two high-way datasets.** **a**, CUB-200-2011, evaluated as 200-way one-shot classification. **b**, Stanford Dogs, evaluated as 120-way one-shot classification. Each point reports mean balanced accuracy and mean task-head runtime across three episodes and 30 support rotations. All methods use the same saved DINOv2 features. The dashed horizontal line is the zero-optimization BL-NCC reference and is not assigned an artificial value on the logarithmic runtime axis. The vertical dash-dotted line marks the 50-ms-per-rotation task-head budget. DCTPR is 3.8 times faster than TIM-ADM on CUB and 49.2 times faster than MAP-RAW on Dogs.

#### 5.4 Cross-episode and cross-dataset stability

The gain over nearest support is consistent across all six episodes (Table 4a). CUB gains range from 6.59 to 6.73 points. Dogs gains range from 5.94 to 6.58 points even though no Dogs label informed method or baseline selection. All 30 Dogs support rotations improve over BL-NCC. The similar gain magnitude across a bird dataset and a dog-breed dataset suggests that the method is correcting a recurring one-shot prototype problem rather than only exploiting one CUB episode construction.

The rotation-level view in Figure 5a strengthens this descriptive result. Every one of the 60 paired rotations improves over BL-NCC, despite different support images and disjoint episode image groups. The dispersion is wider within episodes than the variation among episode means: individual gains range from approximately four to nine points, whereas the six episode means remain between 5.94 and 6.73 points. Because rotations reuse images, these points are evidence of paired stability rather than independent replicates for a significance test.

#### 5.5 Query-prior stress test

The primary protocol supplies the exact class-balanced column mass used by Equation (4). We test the consequence of violating this assumption on CUB. In the mild regime, half the classes retain three queries and half retain nine. In the severe regime, class counts range from one to nine. Counts are permuted across classes with fixed seeds, and macro balanced accuracy prevents majority classes from dominating the metric.

Table 4b shows that uniform-prior DCTPR falls to 68.34% and 68.86%, close to nearest support and below its balanced-query result. Replacing the uniform marginal with true class counts, solely as an oracle diagnostic, recovers 73.35% and 72.87%. The 4.01–5.01 point mean gaps indicate that incorrect class mass, rather than complete loss of query geometry, explains a large fraction of the degradation. Oracle counts are never used as a deployable method input.

Figure 5b places the stress result next to the balanced operating point. The uniform and oracle DCTPR curves coincide when the query set is balanced because the assumed and true class marginals are identical. They separate immediately when the class counts become non-uniform. This pattern localizes the failure to the marginal constraint: query features still contain recoverable information, but the uniform column mass allocates that information to the wrong class totals.

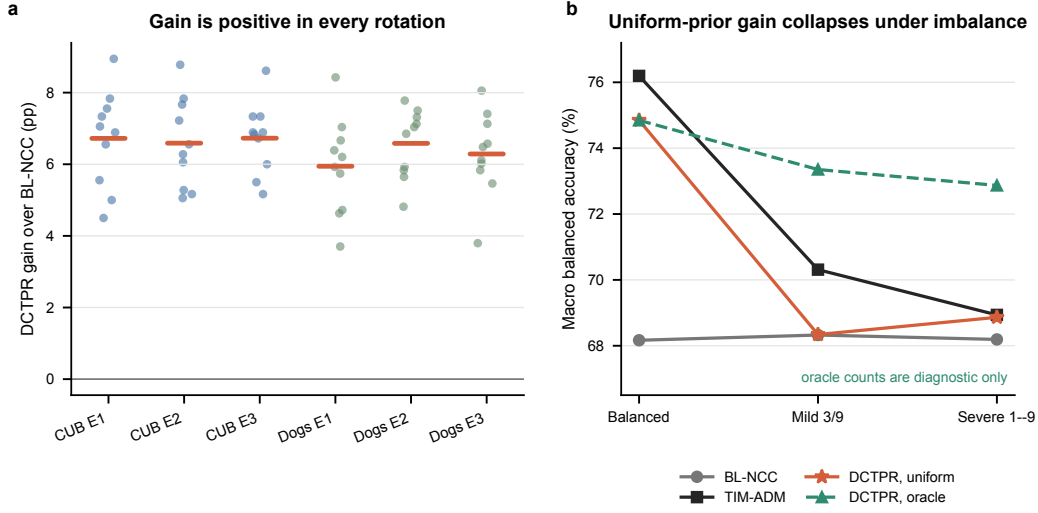


Figure 5. **Stable gains and their query-prior boundary.** **a**, DCTPR gain over BL-NCC for each support rotation; bars mark episode means. All 60 paired differences are positive. **b**, Mean CUB macro balanced accuracy under balanced, mild, and severe query regimes. Uniform-prior DCTPR loses most of its balanced-task advantage under imbalance; true counts are an oracle diagnostic only.

Table 4. **Robustness checks.** **a**, Per-episode stability of the gain over fused nearest support. **b**, CUB macro balanced accuracy under deterministic query imbalance, averaged over three episodes. Oracle counts are diagnostic only.

| <b>a Cross-episode stability</b> |        |       |       | <b>b Query-prior stress test</b>        |              |              |
|----------------------------------|--------|-------|-------|-----------------------------------------|--------------|--------------|
| Episode                          | BL-NCC | DCTPR | Gain  | Method                                  | Mild         | Severe       |
| CUB E1                           | 68.42  | 75.14 | +6.72 | BL nearest support                      | 68.33        | 68.19        |
| CUB E2                           | 68.12  | 74.71 | +6.59 | TIM-ADM                                 | <b>70.31</b> | 68.94        |
| CUB E3                           | 67.96  | 74.69 | +6.73 | DCTPR, uniform prior                    | 68.34        | 68.86        |
| Dogs E1                          | 63.78  | 69.72 | +5.94 | DCTPR, oracle counts                    | 73.35        | <b>72.87</b> |
| Dogs E2                          | 64.46  | 71.05 | +6.58 | Mild: 3/9 queries; severe: 1-9 queries. |              |              |
| Dogs E3                          | 64.04  | 70.32 | +6.29 |                                         |              |              |

## 5.6 Class-level transfer and paired significance

Aggregate balanced accuracy can hide class-level harm, so we decompose the development episodes by class and test the paired decisions directly. This is a post-hoc analysis of the archived per-query predictions: it adds no inference, no tuning, and no official-test access.

Refinement is broadly but not uniformly beneficial (Table 5). On CUB, 152 of 200 classes improve, 20 are unchanged, and 28 degrade; on Dogs, 99 of 120 improve and 19 degrade. Harm is therefore not confined to a handful of outliers, but it is shallow relative to the gain: mean harm among degraded classes is  $-3.47$  and  $-2.71$  points, against mean gains of  $+6.68$  and  $+6.27$  points. The worst single class loses 15.93 points on CUB and 7.78 points on Dogs. No class collapses to zero accuracy under either method, so the update does not sacrifice whole classes to raise the mean. The median class gain ( $+4.63$  and  $+5.37$  points) is below the mean, indicating a right-skewed distribution in which a subset of poorly anchored classes contributes disproportionately to the aggregate improvement.

Paired exact McNemar tests within each support rotation, where the query images are distinct, confirm the direction of both comparisons. Against BL-NCC, all 60 rotations are significant at  $\alpha = 0.05$  and all favour DCTPR. Against the strongest matched solver the tests are informative in the opposite direction and differ by dataset. On CUB, 19 of 30 rotations significantly favour TIM-ADM and none favour DCTPR, so the 1.35-point deficit is a real ordering rather than noise. On Dogs, only 3 of 30 rotations reach significance, all favouring MAP-RAW; the remaining 27 are statistically indistinguishable. The 0.61-point Dogs deficit is thus weakly resolved at this sample size, which strengthens rather than weakens the efficiency argument, since DCTPR reaches that accuracy with about 2% of MAP-RAW’s task-head runtime.

The two solvers also fail on largely the same queries. Error-set Jaccard overlap is 0.78 on both datasets, with only 3.70% (CUB) and 3.87% (Dogs) of queries misclassified by DCTPR alone. Exclusive-error mass is therefore small and nearly symmetric with the strongest solver’s own exclusive errors. This bounds what solver-agreement ensembling could recover in this regime and is consistent with our own abandoned ensemble attempt, which produced a gain far below its preregistered threshold. It also indicates that the residual errors are dominated by shared representation-level ambiguity rather than solver-specific behaviour.

Table 5. **Class-level transfer and paired tests** on the six development episodes (30 support rotations per dataset). Transfer counts compare per-class accuracy against fused nearest support. McNemar tests are two-sided exact tests computed within single rotations and are reported as counts, not pooled, because rotations reuse images.

| Quantity                                    | CUB (200)          | Dogs (120)         |
|---------------------------------------------|--------------------|--------------------|
| Positive / neutral / negative transfer      | 152 / 20 / 28      | 99 / 2 / 19        |
| Mean, median class gain (pp)                | $+6.68$ , $+4.63$  | $+6.27$ , $+5.37$  |
| Mean, worst harm among negatives (pp)       | $-3.47$ , $-15.93$ | $-2.71$ , $-7.78$  |
| Classes at zero accuracy                    | 0                  | 0                  |
| Strongest matched solver                    | TIM-ADM            | MAP-RAW            |
| Classes better / equal / worse              | 53 / 33 / 114      | 43 / 4 / 73        |
| Rotations significant vs. BL-NCC, strongest | 30/30, 19/30       | 30/30, 3/30        |
| Error Jaccard; DCTPR-only, strongest-only   | 0.78; 3.70%, 2.35% | 0.78; 3.87%, 3.26% |

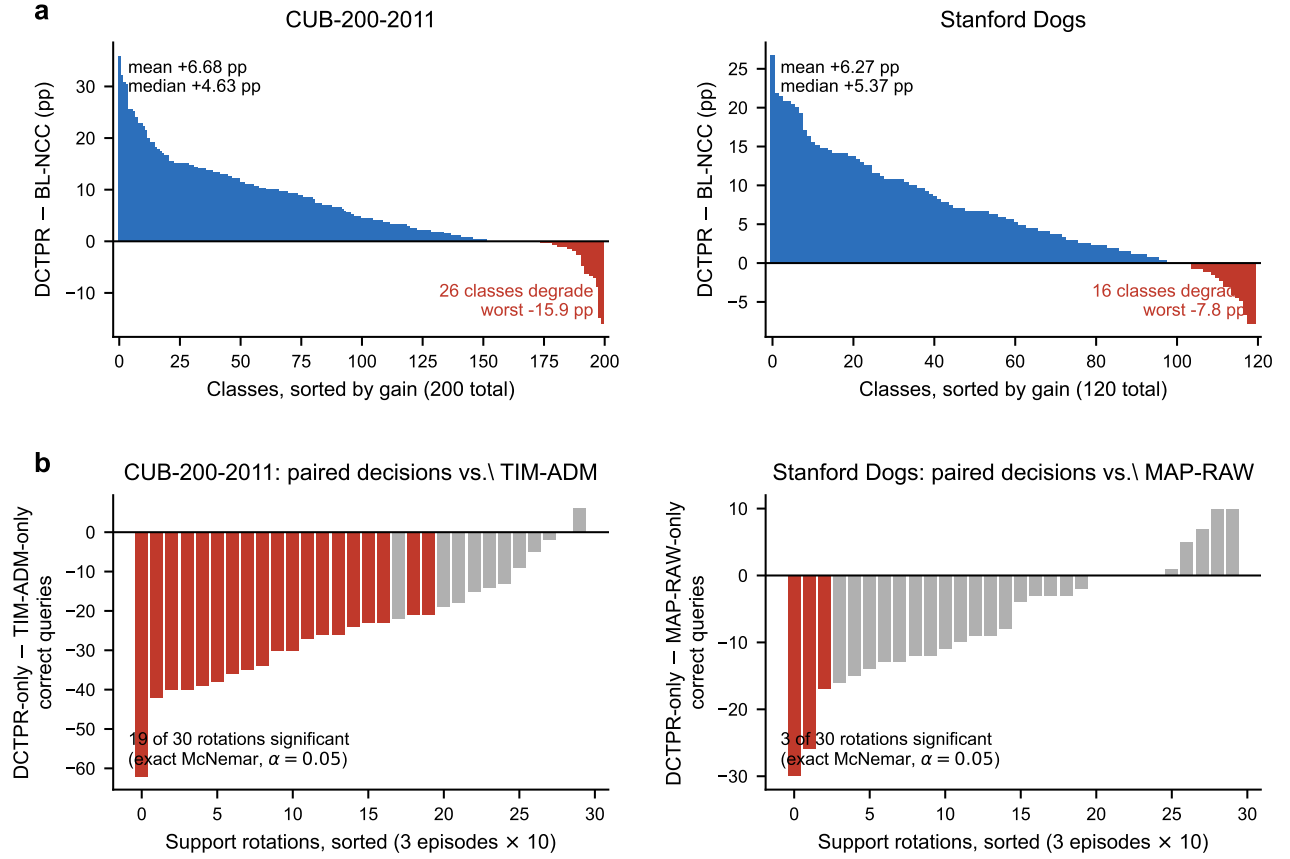


Figure 6. **Class-level and rotation-level paired analysis.** **a**, DCTPR minus BL-NCC balanced accuracy for each class, sorted descending. Blue bars are positive transfer; red bars are classes that degrade. **b**, Per-rotation difference in exclusively-correct query counts between DCTPR and the strongest matched solver (TIM-ADM for CUB, MAP-RAW for Dogs). Red bars reach two-sided exact McNemar significance at  $\alpha = 0.05$ ; grey bars do not. Rotations reuse episode images and are not independent replicates. Source data, hashes, and statistical caveats are archived in [source\\_data/per\\_class\\_transfer\\_summary.json](#).

## 6. DISCUSSION

Short support-anchored refinement recovers 83.3–91.1% of the strongest matched solver’s improvement over nearest support. Within the 50-ms task-head budget, it also gives the highest accuracy among the matched solvers on both datasets, which is a scoped low-latency state-of-the-art result. Component analysis attributes most of the correction to balanced assignment, with dual-capacity refinement adding 1.35 points. This is not a universal Pareto frontier beyond the tested solvers, hardware, and episodes.

TIM-ADM adds 1.35 points on CUB at 3.8 times the task-head runtime; MAP-RAW adds 0.61 points on Dogs at about 49 times the runtime. Maximum-accuracy offline use may therefore prefer those solvers. For one uncached episode, dual-encoder feature extraction dominates total latency. The head-level advantage matters when features are cached or support configurations are evaluated repeatedly. Paired rotation-level tests separate these two deficits: the CUB gap is consistently resolved in TIM-ADM’s favour, whereas 27 of 30 Dogs rotations cannot distinguish DCTPR from MAP-RAW. The accuracy cost of the fast head is therefore dataset-dependent, and on Dogs it is small enough that this protocol does not resolve it reliably.

Class-level decomposition qualifies the aggregate gain. Roughly one class in seven is worse than nearest support, and the worst CUB class loses 15.93 points. The gain is a favourable trade across the class distribution, not a uniform improvement, and a deployment that cannot tolerate per-class regressions would need a class-level safeguard that this method does not provide. The high error-set overlap with the strongest solver further suggests the remaining headroom lies in the frozen representation rather than in the choice of transductive solver.

Fusion improves over both single-capacity variants on every CUB episode and transfers to Dogs without retuning. However, it increases storage and does not identify complementary semantic attributes; dual capacity remains an empirical adapter choice rather than a demonstrated representation mechanism.

The balanced query prior is the main scope limitation: forcing uniform class mass removes most of the advantage under imbalance, consistent with prior evidence.<sup>11</sup> Future work should estimate proportions without query labels and compare on public Dirichlet-imbalance protocols against specialized methods.<sup>17,19–21</sup>

Interpretation is further bounded by matched feature-level reimplementations, image-reusing support rotations, and the known balanced query prior. The locked official-test audit reduces, but does not remove, protocol dependence; public imbalanced evaluation with matched backbones and multi-hardware end-to-end latency remain future work.

## 7. CONCLUSION

We introduced DCTPR, a training-free dual-capacity adapter for balanced, high-way, one-shot fine-grained recognition. Across CUB and Stanford Dogs, it improves fused nearest-support accuracy by 6.68 and 6.27 percentage points, respectively. It retains 83.3–91.1% of the strongest matched solver’s gain while using 2.0–26.4% of that solver’s task-head runtime. A locked official-test audit preserves the same relative behavior without retuning. Within a 50-ms task-head budget, it achieves the highest accuracy among the matched solvers on both datasets, establishing a scoped low-latency state-of-the-art result. These results support short, support-anchored query-center refinement as a practical operating point when features are cached and the query prior is balanced. Removing that prior requires a different inference design.

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